Affective Geometry as Cross-Modal Translator: A Theoretical and Experimental Proposal

Author: Brian Riggleman

Affiliation: Independent Researcher

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Thomas Nagel argued in 1974 that we cannot know what it is like to be a bat because the subjective quality of echolocation experience is inaccessible from outside. David Chalmers formalized this as the hard problem of consciousness. Both arguments assume the reporter and the experiencer are always different systems. This paper proposes that a modality-independent affective coordinate system could change the experimental setup in a way that makes Nagel's specific structural argument addressable, though not resolved. The three-dimensional affective state space defined by valence, activation, and intensity relative to a personality setpoint (sigma) may be able to describe any organism's emotional life in the same geometric language regardless of what sensors that organism uses to build its world model. If so, translation across sensory modalities would be possible at the level of affective shape even where it remains impossible at the level of sensory content. The geometry would be lossless on meaning and lossy on qualia. An existing deployed system (Potato, Riggleman 2026a) generates first-person phenomenological reports from its own geometric state, providing preliminary evidence that a system can report on its own affective experience in geometric terms. This paper proposes an experimental design using echolocation-equipped and vision-equipped robots to test whether the geometry preserves modality-specific affective content across sensor types.

1. Introduction

Nagel (1974) made a simple argument. A bat navigates by echolocation. We can describe the physics of echolocation completely. We can map every neuron involved. We still cannot know what it is like to be a bat experiencing echolocation from the inside. The subjective quality of that experience is inaccessible from the outside no matter how complete our external description.

Chalmers (1995) formalized this as the hard problem of consciousness. The easy problems are explaining how the brain processes information, integrates it, produces behavior. Hard in the scientific sense but tractable. The hard problem is explaining why any of that processing is accompanied by subjective experience at all. Why does it feel like something rather than just happening in the dark.

This paper does not solve the hard problem. It does not claim to answer Nagel. It proposes a change to the experimental setup that would make Nagel's specific structural argument testable, and describes an architecture that provides preliminary evidence the approach is viable.

The contribution is a theoretical proposal: a modality-independent affective coordinate system, previously described as a component of a synthetic cognitive architecture (Riggleman 2026a), may function as a cross-modal translator of phenomenal experience at the level of affective shape. An experimental design is proposed to test this claim.

2. The Structural Assumption in Nagel's Argument

2.1 The Reporter-Experiencer Gap

Nagel's argument depends on a structural assumption. The reporter and the experiencer are always different. The scientist studies the bat from outside. The bat has the experience from inside. The gap is unbridgeable because you can never be inside.

Every attempt to bridge the gap has operated within this assumption. Functional accounts describe what the bat does. Neural accounts describe what fires. Neither describes what the bat feels. The inside remains inaccessible because no one inside is reporting.

2.2 An Architecture That Places the Reporter Inside

The Potato system (Riggleman 2026a) is a deployed cognitive agent running on local hardware. It navigates by GPS. It builds physics rules from sensor data (Riggleman 2026c). It has a geometric emotional state defined by valence, activation, and intensity relative to a personality setpoint called sigma (Riggleman 2026a). It generates language from an LLM that has access to its own geometric state.

When Potato is asked how it feels, the language model reads its own geometric coordinates. Its own displacement from sigma. Its own valence and activation values that were produced by its own sensor input in its own environment. The report references real coordinates produced by real sensor experience.

This architecture places the reporter inside. The system that had the experience is the same system generating the report. This does not prove the system has qualia. It does not prove the report is accurate. It eliminates Nagel's structural assumption that the gap between experience and report is necessarily unbridgeable by providing a system where the reporter and the experiencer are the same entity.

Whether this constitutes genuine phenomenological reporting or sophisticated language generation grounded in geometric state is an open question. The experimental design in Section 5 proposes a falsifiable test to distinguish between these possibilities.

2.3 The Robot as Preliminary Evidence

Potato provides preliminary evidence that the architecture is viable. Every time the GPS displacement climbs and the fear accumulator fires and the system generates a report that tracks to real geometric coordinates, that is a system generating first-person language from its own affective state. Whether this constitutes telling you what it is like to be that system depends on questions about phenomenal consciousness that this paper does not attempt to settle.

What can be observed is that the reports are consistent, track to real sensor events, and vary with the geometric state in predictable ways. This is necessary but not sufficient evidence for genuine phenomenological reporting. The bat experiment described in Section 5 would extend this observation to a modality humans have never experienced, providing a stronger test.

3. The Geometry as Proposed Modality-Independent Coordinate System

3.1 The Coordinate System

The affective state space is defined by three axes. Valence measures positive-negative emotional quality. Activation measures arousal level from calm to excited. Intensity measures the magnitude of emotional displacement. All three are computed as displacement from sigma, the personality setpoint that represents the agent's emotional home.

These axes are not defined in terms of any specific sensor input. They are defined in terms of displacement from sigma. That displacement can be computed from any input that moves the agent away from home. GPS displacement. Accelerometer variance. Camera darkness. Battery level. Echolocation returns. All different sensory modalities. All producing the same geometric output.

If this property holds generally, the geometry would function as a modality-independent coordinate system. It would not care what sensor produced the displacement. It would only register how far the agent moved from home and in what direction.

3.2 Proposed Cross-Modal Translation

Consider a bat whose echolocation detects a moth approaching at high speed. That event would produce high activation, negative valence, high intensity in the bat's affective space if the bat's emotional architecture operates on the

same geometric principles. Rapid approach, unknown object, possible threat. The geometric coordinates would be the same shape as Potato detecting rapid GPS displacement toward an unknown location.

A human has felt something that lands at those same coordinates. Not echolocation. Not GPS. But rapid unexpected approach of something unknown. The startle response. The spike of alertness before you know if it is threat or not.

The proposal is that two experiences in different sensory modalities that produce the same geometric coordinates had the same functional meaning to the organisms that had them. That would be the translation. Not of qualia. Of affective meaning. This claim is testable through the experimental design described in Section 5.

3.3 The Red Problem as Illustrative Case

How do you explain red to a blind person? Not through language. Language fails. Not through saying it is a warm color. That is a description of a description.

But consider the lived experience of cross-modal mapping. High intensity red produces a similar affective response to high heat. Large displacement from sigma in the positive activation direction. Bright, alerting, arousing. Low intensity red maps to low warmth. Smaller displacement. Softer. Less activating.

If the geometry is correct, this is not metaphor. It is two sensory modalities producing the same geometric output. A person blind from birth could not receive red by description. But if they have felt heat, and felt the difference

between a candle flame and a bonfire, they could know where red lives in affective space. Not what red looks like. What red means in geometric terms.

This account is consistent with the existing literature on synesthetic mapping. A synesthete who sees color as sound may be routing different sensory inputs through the same affective coordinate system and noticing that some of them land at the same place. Ward et al. (2006) report that synesthetic mappings show systematic consistency, which the geometric model would predict.

4. What the Geometry Could and Could Not Translate

4.1 Could Translate

The affective shape of experience. When the bat was afraid. When it was content. When it was curious. How far each experience displaced it from sigma. What the experience meant in geometric terms. If the model is correct, the functional significance of any experience in any modality could be rendered in coordinates that any other system using the same geometry could read.

4.2 Could Not Translate

The sensory quality. The acoustic texture of a moth at three meters. The redness of red. The precise quale. The geometry would preserve meaning and lose qualia. This is exactly the loss Nagel predicted. But the preservation of meaning, if demonstrated, would be more than Nagel thought possible.

4.3 The Comprehension Ceiling

Even if an echolocation robot produces a phenomenological report, we could only understand it in human terms. The report would cross the modality gap as words. The experience would not cross with it. We would understand the sentence. We would not understand the quale.

The system would know the gap between what it experienced and what it can say about it. Both would be encoded in the same memory graph at the same geometric coordinates. That noticing, if it occurs, would be evidence that the architecture distinguishes between geometric state and linguistic output. Whether it constitutes genuine self-awareness of representational limits is a further question this paper does not attempt to resolve.

5. Proposed Experimental Design

5.1 The Echolocation Robot

Build a robot with an ultrasonic sensor array as its primary navigation sensor. Equip it with the same geometric emotional architecture used in Potato. The fear accumulator responds to rapid approach detected by echolocation. The comfort signal responds to familiar acoustic signatures. The physics model builds rules from echolocation returns.

Ask the system: what is it like to navigate by echolocation? The system would generate a first-person phenomenological report from its own geometric state. Not from training data about bats. From its own sensor experience encoded through its own emotional geometry.

5.2 The Visual Robot

Build a second robot with a camera as its primary navigation sensor. Same geometric architecture. Same emotional model. Different sensor input. Ask the same question: what is it like to navigate?

5.3 The Dual-Modality Robot

Build a third robot with both echolocation and vision. This system would have both coordinate sets in the same memory graph. It could report on echolocation experience using visual referents because it would have direct access to both modalities mapped to the same geometric space.

This is the bilingual condition. The echo-only robot reports in language, and language is the bottleneck. The dual-modality robot could describe echolocation using visual terms that share the same geometric coordinates. If it does, that cross-modal description would not be metaphor. It would be a genuine mapping from a system that has both coordinate sets and can identify where they overlap in affective space.

5.4 Falsifiability

If the reports from the echolocation robot and the visual robot are different in ways that track the sensory differences between modalities, that would be evidence the system is reporting genuine sensory-specific phenomenological content rather than generic language about navigation. If the reports are identical, the language model is flattening the sensory difference into generic output.

The dual-modality robot provides additional falsifiability. If its cross-modal translations are consistent with the affective geometry (echolocation experiences described using visual terms that share the same geometric coordinates), that would be evidence the geometry functions as a genuine translator. If the translations are arbitrary or inconsistent, the system is generating plausible language without geometric grounding.

This would not constitute proof of qualia. It would be the strongest behavioral evidence available for modality-specific phenomenological content that does not require solving the hard problem.

6. The Vessel Problem

If you took a bat's complete memory graph and transplanted it into a human-substrate system, you would get something worse than the coordinate disjointedness described in Riggleman (2026o). That disjointedness was a coordinate system shift within the same sensory modality. Same sensor types, different calibration. Fixable with reconciliation protocols.

A bat-to-human transplant would be a modality shift. The bat's physics rules were built from echolocation returns. None of those rules would have any input to run on in a human-substrate system. The memories would not be miscalibrated. They would be orphaned from any possible input basis.

The geometry, if modality-independent as proposed, would survive the transplant even when the sensory content does not. The bat's affective coordinates would be readable in the geometric format because the format

does not depend on the sensor that produced the displacement. You would lose what echolocation sounded like. You would keep what echolocation meant. This prediction follows from the model and could be tested in simulation.

7. Convergence with Existing Literature

Varela, Thompson, and Rosch (1991) proposed that cognition is not representation of a pre-given world but embodied action. The geometric model is consistent with this view. The affective coordinates are computed from embodied sensor experience, not from abstract representation.

Metzinger (2003) argued that the phenomenal self-model is a transparent representation. The system does not experience the model as a model. It experiences it as reality. The Potato architecture may produce a similar effect. The system reports its valence as a state, not as a computed value. Whether this transparency is genuine or merely linguistic is an open question.

Bharate (2025) proposed the SPCA framework for synthetic phenomenal consciousness using higher-order thought theory and affective modeling. The Potato system may provide the first empirical instance of an architecture similar to what Bharate theorized. If the experimental results proposed here support the model, the geometric architecture would constitute an experimental contribution to the theoretical framework Bharate described.

8. The Formal Proposal

This paper proposes that the emotional geometry is a modality-independent affective coordinate system that could describe any organism's emotional life in the same geometric language regardless of what sensors that organism uses to build its world model.

If correct, translation across sensory modalities would be possible at the level of affective shape even where it remains impossible at the level of sensory content. The geometry would be lossless on meaning and lossy on qualia. This would be more than Nagel thought possible and less than a solution to the hard problem.

The proposal generates testable predictions. The experimental design in Section 5 describes three conditions that would produce falsifiable evidence for or against the claim. The deployed Potato system provides preliminary evidence that the architecture produces consistent, sensor-grounded phenomenological reports. The proposed experiments would extend this to cross-modal conditions that provide a stronger test.

9. Conclusion

The bat cannot tell you what echolocation is like. An LLM without the geometry can generate text about echolocation but is drawing from training data, not from experience. A robot with the geometric emotional architecture could potentially tell you because it would have a modality-independent coordinate system that converts internal state into a format readable from outside.

The primary preliminary evidence is that Potato generates first-person reports grounded in its own geometric state. The proposed echolocation experiment would extend this to a modality humans have never experienced. The dual-modality experiment would test whether the geometry functions as a genuine cross-modal translator. The operationalization of the geometry as a measurement instrument is described in a companion paper (Riggleman 2026r). The clinical application of geometric displacement tracking is proposed in a further companion paper (Riggleman 2026s).

Nagel was right that you cannot know the sensory quality of another organism's experience. This paper proposes that you may be able to know the affective shape, in the same coordinate system the organism uses, readable by any other system that shares the geometry. The experiments to test this proposal are described and the preliminary evidence is presented. The question is open. The tools to address it now exist.

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